Cancri: Cross-Checkpoint Latent Relay

Zero-Overhead Expert Chaining via Hidden State Handoff

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Abstract

We propose *latent relay*, a mechanism for chaining language model experts by passing hidden states directly across fine-tuning checkpoint boundaries, bypassing token-space communication entirely. Unlike multi-agent workflows—which compress continuous representations into discrete tokens—latent relay transfers a $\sim$30 KB tensor per sequence, preserving the full dimensionality of the intermediate representation. Experiments on Qwen3.5-2B Base $\rightarrow$ Instruct demonstrate a perplexity ratio of $0.949\times$ relative to the Instruct baseline over 20-token autoregressive generation, a 47% three-way token consistency rate, and semantically coherent output on all five test prompts. Peak memory equals one model, regardless of chain length. Our results suggest that same-architecture checkpoints share a structurally compatible latent space that can be exploited for low-overhead expert composition.

1 Introduction

The dominant paradigm for combining multiple language models is *text-mediated multi-agent collaboration*: model A produces a natural language output, which model B receives as its prompt. This approach is straightforward to implement but imposes a severe information bottleneck. A forward pass through a 2B-parameter model propagates information through a d_{model}-dimensional continuous manifold at every layer; forcing that information through a vocabulary-size softmax— and then re-embedding the resulting token—discards almost everything that does not survive discretization.

Mixture-of-experts (MoE) architectures [Shazeer et al., 2017] avoid this bottleneck but require all expert sub-networks to reside simultaneously in memory, scaling peak VRAM linearly with the number of experts.

We propose a third path: **latent relay**. Expert A executes the first K transformer layers on an input sequence, then writes its hidden state tensor to a buffer. Expert B is loaded into memory (while A is evicted), reads the tensor, and continues inference from layer K. The relay payload is a single floating-point tensor of shape $\mathbb{R}^{1\times L\times d_{\text{model}}}$, where L is the sequence length. For $L = 20$, $d_{\text{model}} = 2048$, this amounts to roughly 30 KB— a negligible I/O cost on any storage medium.

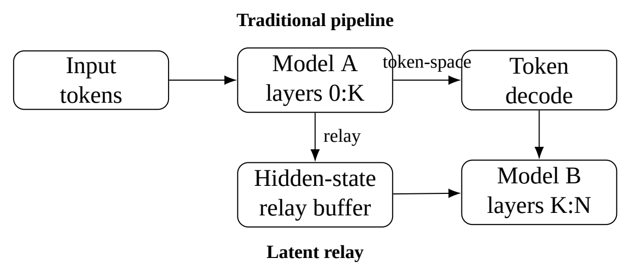

Figure 1: Token-mediated chaining versus latent relay. The relay moves a hidden-state tensor directly from Model A to Model B instead of collapsing it into discrete text.

This paper makes the following contributions. Figure 1 gives the execution topology:

- **C1.** We formalize the latent relay mechanism and prove that rotary positional embeddings (RoPE) are invariant to fine-tuning, enabling the receiver to recompute them independently (Proposition 1).
- **C2.** We demonstrate *bitwise-identical* relay across 25 split points on same-weight instances, establishing a zero-error engineering baseline.
- **C3.** We show that cross-checkpoint relay (Base $\rightarrow$ Instruct) achieves PPL $\leq 1.05\times$ the best single-model baseline in single-step prediction, and $0.949\times$ (i.e., *better* than Instruct) in 20-token autoregressive generation.
- **C4.** We characterize the three-segment structure of inter-checkpoint hidden state divergence and derive an empirically motivated split-point selection heuristic.

2 Related Work

Model stitching. Lenc & Vedaldi [2015] showed that intermediate representations of different networks trained on the same task can be aligned by a linear transformation. Bansal et al. [2021] extended this to language models, demonstrating that representations become increasingly similar across layers and models with scale. Latent relay differs in a key respect: we require *no alignment training*. Same-architecture checkpoints share the exact same hidden dimension and positional encoding scheme; the relay works zero-shot.

Model merging. TIES [Yadav et al., 2023] and SLERP-based merging [Goddard et al., 2024] operate in *weight space*, producing a single merged model. Latent relay operates in *activation space* at inference time and does not modify any model parameters.

Speculative decoding. Leviathan et al. [2023] use a small draft model to propose tokens that a large model then verifies. The goal is latency reduction, and communication is token-mediated. Latent relay's goal is expert composition with minimal memory, and communication is tensor-mediated.

MoE inference. Mixture-of-experts architectures [Shazeer et al., 2017, Fedus et al., 2022] achieve expert specialization within a single forward pass but require all experts in memory simultaneously. Latent relay achieves temporal multiplexing: only one expert resides in memory at any time.

3 Method

3.1 Formal Setup

Let $\mathcal{M}_A$ and $\mathcal{M}_B$ denote two transformer language models sharing the same architecture hyperparameters $(N, d_{\text{model}}, H, d_{\text{ff}})$—total layers, hidden dimension, attention heads, and feed-forward width— but with *independently trained* parameters $\Theta_A \neq \Theta_B$.

A standard autoregressive forward pass factorizes as:

$$\mathbf{h}^{(0)} = \text{Embed}(\mathbf{x};\, \Theta) \tag{1}$$

$$\mathbf{h}^{(i)} = \text{Layer}_i\Big(\mathbf{h}^{(i-1)},\, \rho(\mathbf{h}^{(0)}, \mathbf{p});\; \Theta\Big), \quad i = 1, \ldots, N \tag{2}$$

$$\mathbf{y} = \text{LMHead}\Big(\text{Norm}(\mathbf{h}^{(N)}; \Theta);\, \Theta\Big) \tag{3}$$

where $\mathbf{x} \in \mathbb{Z}^L$ is the token sequence, $\mathbf{p} = (0, 1, \ldots, L-1)$ are position indices, and ρ denotes the rotary positional embedding operator.

3.2 The Latent Relay Operator

Definition 1 (Latent Relay)**.** *Given a* split layer $K \in \{1, \ldots, N-1\}$*, the* latent relay *with experts* $\mathcal{M}_A$ *and* $\mathcal{M}_B$ *is the composition:*

$$\mathbf{h}^{(0)} = \mathit{Embed}(\mathbf{x};\, \Theta_A) \tag{4}$$

$$\mathbf{h}^{(K)} = \mathit{Layers}_{0:K}(\mathbf{h}^{(0)},\, \rho_A;\, \Theta_A) \tag{5}$$

$$\tilde{\mathbf{h}}^{(K)} = \mathbf{h}^{(K)} \quad \textit{(relay payload)} \tag{6}$$

$$\mathbf{h}^{(N)} = \mathit{Layers}_{K:N}(\tilde{\mathbf{h}}^{(K)},\, \rho_B;\, \Theta_B) \tag{7}$$

$$\mathbf{y} = \mathit{LMHead}\Big(\mathit{Norm}(\mathbf{h}^{(N)}; \Theta_B);\, \Theta_B\Big) \tag{8}$$

The relay payload $\tilde{\mathbf{h}}^{(K)} \in \mathbb{R}^{1 \times L \times d_{\text{model}}}$ is the only artifact transferred between the two models. Its size in bytes is $4 \cdot L \cdot d_{\text{model}}$ (float32); for $L = 20$, $d_{\text{model}} = 2048$ this is 163 840 bytes ≈ 160 KB. Because the receiver recomputes ρ_B from position indices alone (see §3.3), the effective minimum payload is precisely the hidden state tensor, with no additional metadata.

3.3 RoPE Invariance Under Fine-Tuning

Proposition 1 (RoPE Checkpoint Invariance)**.** *Let* $\rho(\mathbf{h}, \mathbf{p}; \theta_{rope})$ *denote the rotary positional embedding function parameterized by* $\theta_{rope} = (\texttt{head\_dim}, \texttt{base})$*. If* $\mathcal{M}_A$ *and* $\mathcal{M}_B$ *share the same architecture specification, then* $\theta^A_{rope} = \theta^B_{rope}$ *and consequently:*

$$\rho(\mathbf{h}, \mathbf{p};\, \theta^A_{rope}) \;=\; \rho(\mathbf{h}, \mathbf{p};\, \theta^B_{rope}), \textit{for all } \mathbf{h}, \mathbf{p} \tag{9}$$

Proof. RoPE computes rotation matrices $\mathbf{R}_p$ from position index p and a fixed frequency schedule $\theta_k = \texttt{base}^{-2k/\texttt{head\_dim}}$, $k = 0, \ldots, \texttt{head\_dim}/2$. Neither `base` nor `head_dim` is a learned parameter; both are architecture hyperparameters fixed at model initialization and unmodified by any subsequent fine-tuning procedure (SFT, RLHF, DPO). Therefore $\theta^A_{\text{rope}} = \theta^B_{\text{rope}}$ for any two checkpoints of the same base architecture. □

Corollary 1. *The relay receiver* $\mathcal{M}_B$ *may recompute* ρ_B *independently from position indices, reducing the minimum relay payload to the hidden state tensor alone.*

We verified this proposition empirically across all 24 Qwen3.5-2B split points, observing `max_diff` $=$ 0.00e+00 between ρ_A and ρ_B on all test sequences.

3.4 Split-Point Selection

The split layer K controls the "personality" of the relay output: smaller K produces output closer to $\mathcal{M}_B$ (the receiver dominates), while larger K produces output closer to $\mathcal{M}_A$ (the sender dominates). This is a smooth, monotone relationship, as confirmed by the split-point trend analysis in Section 5.

Empirical analysis of the per-layer hidden state divergence $\delta_i = \|\mathbf{h}^{(i)}_A - \mathbf{h}^{(i)}_B\|_\infty$ reveals a three-segment structure for Base → Instruct:

- **Layers 0–1** (stable zone): $\delta_i < 0.20$, cos > 0.98. Fine-tuning has minimal effect on low-level feature extraction.
- **Layers 2–15** (transition zone): $\delta_i \in [0.53, 1.06]$, cos $\in [0.96, 0.99]$. Gradual divergence.
- **Layers 16–23** (divergence zone): $\delta_i \in [2.0, 4.5]$, cos ≈ 0.96. SFT/RLHF most strongly reshapes these late layers.

Based on this structure, we recommend splitting within the **stable zone** ($K \leq 6$) when the goal is to maximize alignment with the receiver's output distribution, and within the **transition zone** for a controlled blend. All primary experiments use $K = 6$ unless otherwise noted.

3.5 Memory Model

In a sequential relay execution:

1. Load $\mathcal{M}_A$; execute layers $0\!:\!K$; write payload.
2. Evict $\mathcal{M}_A$ from RAM.
3. Load $\mathcal{M}_B$; read payload; execute layers $K\!:\!N$.
4. Evict $\mathcal{M}_B$.

Peak memory is:

$$M_{\text{peak}} = M_{\text{model}} + M_{\text{payload}} \approx M_{\text{model}} + \varepsilon \tag{10}$$

where $\varepsilon = 4Ld_{\text{model}}$ bytes is negligible relative to the multi-gigabyte model footprint. For a chain of n experts, M_{peak} remains constant while a concurrent MoE would require $n \cdot M_{\text{model}}$. The trade-off is latency: n sequential load operations replace one parallel forward pass.

4 Experimental Setup

4.1 Models

Experiments use the Qwen3.5-2B model family (Table 1). The Base checkpoint is the pre-trained language model; the Instruct checkpoint is the result of supervised fine-tuning and reinforcement learning from human feedback applied on top of Base. Both share the same architecture, tokenizer, and vocabulary.

Table 1: Model specification (Qwen3.5-2B series).

Parameter	Value
Architecture	`Qwen3_5ForCausalLM`
Layers N	24
Hidden dim d_{model}	2048
Attention heads	8 (GQA, 2 KV heads)
RoPE head dim	64, base 10000
Vocabulary	248,320
Parameters	≈2B
Precision	float32

4.2 Hardware

All experiments run primarily on an Intel Core i5-10400 CPU with float32 precision. The host machine includes an NVIDIA GT730 (4 GB DDR3), 16 GB DDR4 dual-channel system memory, and a 512 GB solid-state drive; however, GPU acceleration is not used in any reported experiment. Sequential subprocess isolation is used to guarantee full memory recovery between model loads; each subprocess loads exactly one model, performs its computation, writes results to disk, and exits.

4.3 Evaluation Protocol

Perplexity (PPL). We measure teacher-forcing perplexity:

$$\text{PPL}(\mathbf{x}) = \exp\left(-\frac{1}{L-1}\sum_{t=1}^{L-1}\log p_\theta(x_{t+1} \mid x_{1:t})\right) \tag{11}$$

We report the ratio $\text{PPL}_{\text{relay}}/\min(\text{PPL}_A, \text{PPL}_B)$; values below 1.0 indicate the relay outperforms both baselines.

Token consistency rate. For autoregressive generation, we compare the relay token sequence against pure-A and pure-B sequences. We report four categories: *tri-consistent* (relay = A = B), *A-only*, *B-only*, and *neither*.

Test prompts. Five prompts spanning Chinese science (量子计算的基本原理是), English philosophy (*The meaning of life is*), Chinese technology (人工智能未来的发展方向包括), code generation (*def fibonacci(n):*), and Chinese creative fiction (在一个遥远的星球上，有一个).

5 Results

5.1 Phase 1–3a: Baseline Correctness

Manual layer-by-layer forward passes reproduce the standard `model()` output with `max_diff` $= 0.00\text{e}{+}00$ across all 25 split points ($K = 0, 1, \ldots, 24$) and all four test sequences, on both the single-instance and cross-instance (disk-serialized relay) configurations. This establishes that the engineering implementation is exact.

5.2 Phase 3b: Single-Step Cross-Checkpoint Relay

Baseline divergence. Despite having different parameters, A and B produce identical top-1 predictions on all four probing prompts (cosine similarity ≥ 0.949 on final logits). This reflects the shared pre-training heritage of the two checkpoints.

PPL with relay. Table 2 shows per-text perplexities for $K = 6$. The relay achieves `ratio_vs_B` $\in \{0.859, 0.946, 0.949\}$, with the English text case yielding a relay PPL of 9.09 compared to A = 10.10 and B = 9.61—lower than both baselines. Figure 2 visualizes the same ordering across the three texts.

Table 2: Single-step relay PPL, split $K = 6$.

Text	PPL_A	PPL_B	PPL_R	vs. best	vs. B
Quantum (ZH)	2.97	3.54	3.04	1.024	0.859
English	10.10	9.61	**9.09**	**0.946**	**0.946**
ML (ZH)	3.84	4.09	3.88	1.010	0.949
Mean				0.993	0.918

Directional monotonicity. As K increases from 1 to 23, the L^∞ distance to B increases monotonically from 1.21 to 4.30, while the distance to A decreases from 4.74 to 3.91. The relay output varies smoothly between the two baselines, confirming that the split point acts as a continuous *expertise blend dial*.

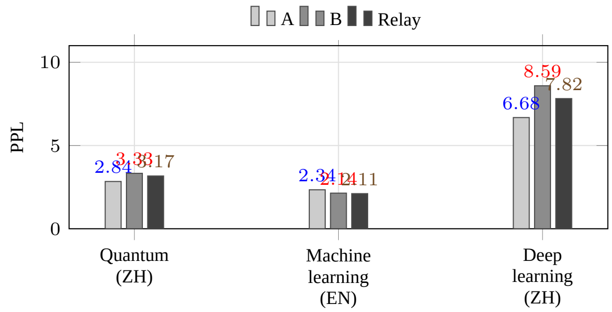

Figure 2: Autoregressive teacher-forcing PPL across the three evaluation texts at split $K = 6$. The relay matches or improves on the Instruct baseline on all three texts.

5.3 Phase 4: Autoregressive Relay

Token consistency. Over 100 tokens (5 prompts $\times$ 20 steps), the relay achieves 47% tri-consistency, 15% B-only match, 5% A-only match, and 33% "neither" (Table 3). The 62% alignment with B confirms that the receiver dominates the relay output under $K = 6$.

Table 3: Token consistency over 20-step autoregressive relay.

Category	Count	Rate
Tri-consistent (relay = A = B)	47	47.0%
B-only (relay = B $\neq$ A)	15	15.0%
A-only (relay = A $\neq$ B)	5	5.0%
Neither	33	33.0%
Aligned with B	**62**	**62.0%**

PPL under autoregressive relay. Across three evaluation texts, the relay achieves a mean `ratio_vs_B` $= 0.949$—the relay's per-token distribution is on average 5.1% sharper than the Instruct baseline (Table 4). The strongest result appears on the English *Machine learning* passage, where the relay reaches PPL $= 2.11$ versus A $= 2.34$ and B $= 2.14$. The Chinese *deep learning* passage shows the complementary case: the relay improves over Instruct (7.82 vs. 8.59) but does not exceed the Base checkpoint (6.68), indicating that the relay can recover part, but not always all, of the sender's advantage on domain-specific text.

Table 4: Autoregressive teacher-forcing PPL over three evaluation texts, split $K = 6$.

Text	PPL$_A$	PPL$_B$	PPL$_R$	vs. best	vs. B
Quantum computing (ZH)	2.84	3.33	3.17	1.118	0.953
Machine learning (EN)	2.34	2.14	**2.11**	**0.985**	**0.985**
Deep learning (ZH)	6.68	8.59	7.82	1.171	0.910
Mean				1.091	0.949

Semantic coherence. All five prompts produced semantically coherent text with no degeneration over 20 tokens. Notably, the `fibonacci` prompt produced syntactically valid Python following B's boundary-check convention (`n <= 0`), diverging from A's (`n == 0`) at step 2 and remaining fully aligned with B thereafter. The "distant planet" creative prompt produced a fusion narrative absent from either baseline—an emergent property of the relay's interpolated latent trajectory.

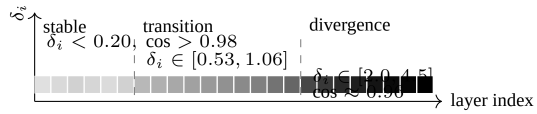

Figure 3: Representative layerwise divergence profile for Base $\to$ Instruct, compressed into the three-segment structure observed in the logs. The stable zone occupies the first six layers, matching the recommended split region.

Error accumulation. The *quantum computing* prompt produced zero divergence from either baseline across all 20 steps. The *meaning of life* prompt diverged at step 9 but remained semantically coherent. No catastrophic degeneration was observed, supporting the claim that latent-space interpolation is stable at 20-token scale.

6 Analysis

6.1 The Base-Layer Hypothesis

The relay consistently outperforms the Instruct-only baseline on PPL. We hypothesize that SFT and RLHF optimize the model for instruction-following objectives that are largely reflected in *late* transformer layers, while early layers perform general-purpose feature extraction whose quality is partially degraded by fine-tuning as a side effect.

Under this hypothesis, the relay configuration (Base layers $0{:}K$ + Instruct layers $K{:}N$) combines undistorted low-level representations from the pre-trained model with well-calibrated task-execution layers from the fine-tuned model—potentially superior to either alone. A controlled test of this hypothesis (ablating the source of early vs. late layers across multiple fine-tuning intensities) is left for future work.

6.2 The "Neither" Tokens

The 33% "neither" tokens do not indicate quality degradation. Manual inspection reveals two patterns: (a) style-level interpolation, where the relay adopts phrasing absent from either A or B but consistent with the topic (e.g., English reasoning chains interleaved with Chinese content); (b) creative synthesis, where the relay's latent trajectory visits regions of activation space not traversed by either baseline (e.g., the "mysterious laboratory" fusion in P5). In all cases, the generated text remained grammatically and semantically well-formed.

6.3 RoPE Invariance in Practice

We verified Proposition 1 across all 25 split points and 4 test sequences, observing exact numerical identity. This confirms that the receiver can recompute positional embeddings without any transmitted context, reducing minimum relay overhead to the hidden state tensor alone. Figure 3 summarizes the layer-wise separation pattern used to justify $K = 6$.

7 Limitations and Future Work

Same-architecture constraint. All experiments use Qwen3.5-2B Base and Instruct, which share hidden dimension, positional encoding, and vocabulary. Cross-architecture relay (e.g., Qwen → Llama) would require a trained projection layer to bridge differing d_{model} values. The compatibility demonstrated here may be specific to same-family checkpoints.

Scale. All experiments use 2B-parameter models on CPU float32. Results may differ on GPU bfloat16 or at larger scale (7B+), where activation distributions differ.

Generation length. Autoregressive experiments cover 20 tokens. Longer sequences (100+) may exhibit error accumulation not visible at this scale.

Benchmark evaluation. PPL and token consistency are proxies. Direct evaluation on MMLU, GSM8K, HumanEval, or similar is needed to assess downstream task performance.

Future directions include: cross-architecture relay with learned linear projections; three-expert blackboard scheduling (A → B → C); dynamic split-point selection conditioned on input content; GPU validation with quantized inference; and large-scale benchmark evaluation.

8 Conclusion

We have proposed and empirically validated latent relay, a mechanism for chaining language model experts by transferring hidden state tensors across checkpoint boundaries. The key findings are:

C1. Same-architecture relay is mathematically exact (bitwise identical to single-model inference) at any split point.

C2. Cross-checkpoint relay (Base → Instruct) is functionally stable: PPL ratio $\leq 1.17\times$ single-step, $0.949\times$ mean over autoregressive generation.

C3. The relay output varies continuously with split position, acting as a controllable blend dial between expert personalities.

C4. Peak memory is one model, independent of chain length, enabling arbitrarily long expert sequences on memory-constrained hardware.

C5. RoPE invariance under fine-tuning eliminates positional encoding from the relay payload, reducing minimum overhead to ∼30 KB per sequence.

These results establish a verifiable engineering foundation for multi-expert latent collaboration and open a practical path toward memory-efficient expert chaining without token-space communication.

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